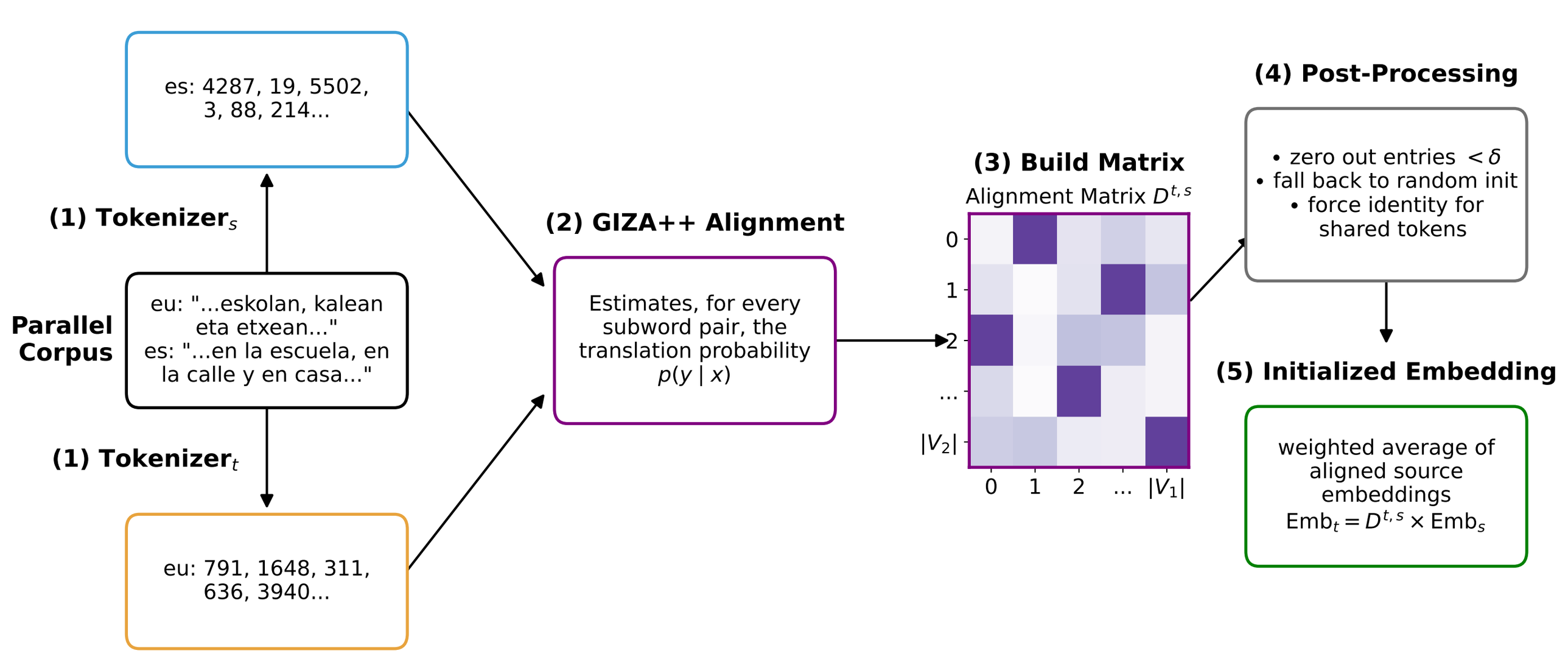
es: 4287, 19, 5502, 3, 88, 214...
(1) Tokenizer$_s$
Parallel Corpus
eu: "...eskolan, kalean eta etxean..." es: "...en la escuela, en la calle y en casa..."
(1) Tokenizer$_t$
eu: 791, 1648, 311, 636, 3940...
(2) GIZA++ Alignment
Estimates, for every subword pair, the translation probability $p(y \mid x)$
(3) Build Matrix
Alignment Matrix $D^{t,s}$
0
1
2
...
$|V_2|$
0
1
2
...
$|V_1|$
(4) Post-Processing
• zero out entries $< \delta$
• fall back to random init
• force identity for shared tokens
(5) Initialized Embedding
weighted average of aligned source embeddings
$\text{Emb}_t = D^{t,s} \times \text{Emb}_s$